

Dynamic Programming Questions

Question 1 : **EASY**

Tribonacci Sequence

The Tribonacci sequence T_n is defined as follows:

$T_0 = 0, T_1 = 1, T_2 = 1$, and $T_{n+3} = T_n + T_{n+1} + T_{n+2}$ for $n \geq 0$.

Given n , return the value of T_n . [\[Go to Qs\]](#)

Question 2 : **MEDIUM**

Maximum profit after buying and selling stocks with transaction fees

You are given an array of prices where $prices[i]$ is the price of a given stock on the i th day, and an integer fee representing a transaction fee.

Find the maximum profit you can achieve. You may complete as many transactions as you like, but you need to pay the transaction fee for each transaction.

Note: You may not engage in multiple transactions simultaneously (i.e., you must sell the stock before you buy again).

The transaction fee is only charged once for each stock purchase and sale. [\[Go to Qs\]](#)

Question 3 : **HARD**

Longest Increasing Path in Matrix

Given an $m \times n$ integers matrix, return the length of the longest increasing path in the matrix.

From each cell, you can either move in four directions: left, right, up, or down. You may not move diagonally or move outside the boundary (i.e., wrap-around is not allowed). [\[Go to Qs\]](#)

Question 4 : **MEDIUM**

Generate Parentheses

Given n pairs of parentheses, write a function to generate all combinations of well-formed parentheses. [\[Go to Qs\]](#)

Question 5 : MEDIUM**House Thief**

You are a professional robber planning to rob houses along a street. Each house has a certain amount of money stashed, the only constraint stopping you from robbing each of them is that adjacent houses have security systems connected and it will automatically contact the police if two adjacent houses were broken into on the same night.

Given an integer array `nums` representing the amount of money of each house, return the maximum amount of money you can rob tonight without alerting the police. [\[Go to Qs\]](#)

Question 6 : MEDIUM**Longest Palindromic Subsequence**

Given a string `s`, find the longest palindromic subsequence's length in `s`.

A subsequence is a sequence that can be derived from another sequence by deleting some or no elements without changing the order of the remaining elements. [\[Go to Qs\]](#)

Question 7 : MEDIUM**Equal Subset Sum Difference**

Given an integer array `nums`, return `true` if you can partition the array into two subsets such that the sum of the elements in both subsets is equal or `false` otherwise. [\[Go to Qs\]](#)

Question 8 : HARD**Mountain Array(Longest Bitonic Subsequence)**

You may recall that an array `arr` is a mountain array if and only if:

- `arr.length >= 3`
- There exists some index `i` (0-indexed) with $0 < i < arr.length - 1$ such that:
 - `arr[0] < arr[1] < ... < arr[i - 1] < arr[i]`
 - `arr[i] > arr[i + 1] > ... > arr[arr.length - 1]`

Given an integer array `nums`, return the minimum number of elements to remove to make `nums` a mountain array. [\[Go to Qs\]](#)

Question 9 : HARD**Box Stacking**

Given `n` cuboids where the dimensions of the `i`th cuboid is `cuboids[i] = [widthi, lengthi, heighti]` (0-indexed). Choose a subset of cuboids and place them on each other.

You can place cuboid i on cuboid j if $\text{width}_i \leq \text{width}_j$ and $\text{length}_i \leq \text{length}_j$ and $\text{height}_i \leq \text{height}_j$. You can rearrange any cuboid's dimensions by rotating it to put it on another cuboid.

Return the maximum height of the stacked cuboids. [\[Go to Qs\]](#)

Question 10 : **MEDIUM**

Palindrome Partitioning

Given a string s , partition s such that every substring of the partition is a palindrome. Return all possible palindrome partitioning of s . [\[Go to Qs\]](#)

